Assessment 3 — Event Management & Workforce Analytics

Environment: PyCharm
Suggested duration: 3 Hours

Create project:

```
python_assessment_3
```

Data File 1 — `registrations.csv`

```
registration_id,event_id,attendee_name,city,tickets,registration_date,sta
R1001,E101,Aarav Shah,Hyderabad,2,2026-07-01,Confirmed,5000
R1002,E102,Simran Kaur,Mumbai,1,2026-07-02,Confirmed,3500
R1003,E101,Faiz Ahmed,Pune,3,2026-07-04,Pending,7500
R1004,E103,Keerthi Rao,Bangalore,2,2026-07-07,Confirmed,6000
R1005,E104,Mohit Kapoor,Delhi,1,2026-07-10,Cancelled,4000
R1006,E102,Ananya Das,Hyderabad,2,2026-08-01,Confirmed,7000
R1007,E103,Zubin Patel,Mumbai,1,2026-08-04,Pending,3000
R1008,E101,Sonal Mehta,Delhi,4,2026-08-09,Confirmed,10000
R1009,E104,Harish Nair,Chennai,2,2026-08-15,Confirmed,8000
R1010,E102,Nida Khan,Hyderabad,3,2026-09-02,Confirmed,10500
R1011,E103,Rohit Bansal,Pune,2,2026-09-05,Cancelled,6000
R1012,E104,Maya Iyer,Bangalore,1,2026-09-11,Confirmed,4000
```

Data File 2 — `events.csv`

```
event_id,event_name,event_type,venue_city,capacity,organizer
E101,AI Innovation Summit,Technology,Hyderabad,500,Vision Events
E102,Leadership Forum,Business,Mumbai,350,Prime Conferences
E103,Data Engineering Bootcamp,Technology,Bangalore,200,SkillWorks
E104,Startup Connect,Business,Delhi,300,Founders Circle
E105,Design Expo,Design,Pune,250,Creative Hub
```

Data File 3 — `staff.json`

```
[
  {
    "staff_id": 501,
    "name": "Arvind Rao",
    "department": "Operations",
    "salary": 72000,
    "skills": ["Coordination", "Excel", "Python"],
    "performance": {
      "rating": 4.3,
      "events_handled": 8
    }
  },
  {
    "staff_id": 502,
    "name": "Sanya Gupta",
    "department": "Marketing",
    "salary": 68000,
    "skills": ["Digital Marketing", "Analytics"],
    "performance": {
      "rating": 4.6,
      "events_handled": 10
    }
  },
  {
    "staff_id": 503,
    "name": "Bilal Ahmed",
    "department": "Technology",
    "salary": 90000,
    "skills": ["Python", "SQL", "Automation"],
    "performance": {
      "rating": 4.8,
      "events_handled": 12
    }
  },
  {
    "staff_id": 504,
    "name": "Trisha Menon",
    "department": "Operations",
    "salary": 76000,
    "skills": ["Coordination", "Vendor Management"],
```

```
    "performance": {
      "rating": 4.1,
      "events_handled": 7
    }
  }
]
```

Section A — Python Basics

1. Take attendee name, number of tickets and ticket price from the user.
2. Calculate total booking amount.
3. Apply:
    - 10% discount when amount is ₹10,000 or more
    - 5% otherwise
4. Display all results using f-strings.
5. Ask the user for event capacity and registered participants.
6. Display:
    - `Full` if registrations equal capacity
    - `Seats Available` if below capacity
    - `Invalid Data` if registrations exceed capacity
7. Print numbers 1–15 using a `for` loop.
8. Display only numbers divisible by 3.
9. Use a `while` loop to count down from 10 to 1.
10. Use `break` and `continue` in suitable examples.

Section B — Lists, Tuples, Sets and Dictionaries

Use:

```
booking_amounts = [
    5000, 3500, 7500, 6000,
    4000, 7000, 3000, 10000
]
```

11. Find total booking amount.
12. Find average booking amount.
13. Find highest and lowest amounts.
14. Display values greater than `5000`.
15. Sort amounts descending.

Use:

```
cities = [
    "Hyderabad",
    "Mumbai",
    "Pune",
    "Hyderabad",
    "Delhi",
    "Mumbai"
]
```

16. Find unique cities using a set.
17. Add another city to the set.
18. Find the number of unique cities.
19. Create a tuple containing:

```
Event ID
Event Name
Venue
Capacity
```

20. Unpack the tuple into separate variables.
21. Create a dictionary representing one registration.

Include:

```
registration_id
attendee_name
city
tickets
status
amount
```

22. Change its status.
23. Increase ticket count.
24. Add an email field.
25. Loop through all dictionary items.

Section C — Functions, Lambda, Filter, Map

26. Create a function:

```
calculate_booking(ticket_price, tickets)
```

27. Create a function that accepts booking amount and returns the discount amount.
28. Create a function that accepts registration status and returns whether the registration is active.
29. Create a function accepting a list of booking amounts and returning the total.
30. Write a lambda function that increases an amount by 10%.
31. Use `filter()` and lambda to select amounts greater than `6000`.
32. Create a list of registration dictionaries and use `filter()` to return Confirmed registrations.
33. Use `map()` to extract attendee cities.
34. Use `map()` and `set()` to obtain unique cities.
35. Sort a list of registration dictionaries using:

```
key=
```

based on amount.

Section D — OOP

Create:

```
Event
```

with:

```
event_id
event_name
capacity
ticket_price
```

36. Use `__init__()` to initialize attributes.

37. Create:

```
display_event()
```

38. Create:

```
calculate_revenue(tickets_sold)
```

39. Create two Event objects.

Create:

```
OnlineEvent
```

that inherits from `Event`.

40. Add:

```
platform
```

41. Use `super()`.

42. Override `display_event()` to also display the platform.

Create:

```
PhysicalEvent
```

43. Add:

```
venue
```

44. Override `display_event()`.

45. Put OnlineEvent and PhysicalEvent objects into a list.

46. Call `display_event()` for every object to demonstrate polymorphism.

Section E — Text File Handling

Create:

`event_log.txt`

47. Write three event-processing messages.

48. Append two more messages.

49. Read the complete file.

50. Read the file line by line.

51. Display every line without additional blank lines.

52. Use `with open()` rather than manually closing the file.

Section F — CSV Processing Without Pandas

Using:

`registrations.csv`

53. Read the file using the `csv` module.

54. Store records in a Python list.

55. Display only Confirmed registrations.

56. Display registrations from Hyderabad.

57. Convert `amount` into a numeric datatype.

58. Find registrations where amount exceeds `₹7,000`.

59. Calculate total registration amount.

60. Extract all cities and display unique cities.

Section G — JSON Processing

Using:

```
staff.json
```

61. Read JSON into Python.

62. Check the datatype of the returned object.

63. Display all staff names.

64. Display employees from Operations.

65. Display employees whose salary exceeds ₹75,000.

66. Find employees having Python in their skills.

67. Display:

```
Name
Rating
Events Handled
```

68. Find staff whose performance rating exceeds `4.5`.

69. Extract all skills into one list.

70. Find unique skills.

71. Sort employees by salary descending.

72. Sort employees by performance rating descending.

73. Add `"Pandas"` to staff ID `503`.

74. Save the modified JSON as:

`staff_updated.json`

Section H — User-Defined Modules

Create:

`event_utils.py`

75. Add functions for:

- calculate booking amount
- calculate discount
- check event capacity

Create:

`main.py`

76. Import the entire module.

77. Call all three functions.

78. Import one function directly.

79. Import the module using an alias.

Section I — Pandas Basics

Using `registrations.csv` :

80. Read the file into a DataFrame.

81. Display:

- first 5 rows
- last 5 rows
- shape
- columns

- datatypes

82. Display only:

```
attendee_name
city
status
amount
```

83. Use `query()` to find Confirmed registrations.

84. Find registrations whose amount exceeds ₹6,000.

85. Find Confirmed registrations from Hyderabad.

86. Count registrations by status.

87. Find total amount by status.

88. Find average amount by city.
